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Set 10 Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart

elements.

2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

3. Build a table to show the survey response data for each survey. Label the table elements.

4. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for

interactive exploration of survey responses.

CODE:

```
import matplotlib.pyplot as plt
```

```
survey_id = [1, 2, 3]
```

```
q1 = ['A', 'B', 'C']
```

```
q2 = ['B', 'A', 'A']
```

```
q3 = ['C', 'D', 'B']
```

1. Grouped Bar Chart - Question 1

```
options = ['A', 'B', 'C']  
counts = [q1.count('A'), q1.count('B'), q1.count('C')]
```

```
plt.figure(figsize=(7, 5))  
plt.bar(options, counts)  
plt.xlabel("Answer Options")  
plt.ylabel("Number of Responses")  
plt.title("Question 1 - Distribution of Answers")  
plt.show()
```

2. Stacked Bar Chart - All Questions

```
questions = ['Question 1', 'Question 2', 'Question 3']
```

```
A = [q1.count('A'), q2.count('A'), q3.count('A')]  
B = [q1.count('B'), q2.count('B'), q3.count('B')]  
C = [q1.count('C'), q2.count('C'), q3.count('C')]  
D = [q1.count('D'), q2.count('D'), q3.count('D')]
```

```
plt.figure(figsize=(8, 5))
```

```
plt.bar(questions, A, label='A')  
plt.bar(questions, B, bottom=A, label='B')
```

```
bottom_B = [A[i] + B[i] for i in range(3)]  
plt.bar(questions, C, bottom=bottom_B, label='C')
```

```
bottom_C = [A[i] + B[i] + C[i] for i in range(3)]

plt.bar(questions, D, bottom=bottom_C, label='D')


plt.xlabel("Questions")

plt.ylabel("Number of Responses")

plt.title("Overall Distribution of Survey Responses")

plt.legend()

plt.show()

# 3. Survey Response Table

print("\nSURVEY RESPONSE TABLE")

print("-" * 55)

print("Survey ID\tQuestion 1\tQuestion 2\tQuestion 3")

print("-" * 55)


for i in range(3):

    print(

        survey_id[i],

        "\t\t",

        q1[i],

        "\t\t",

        q2[i],

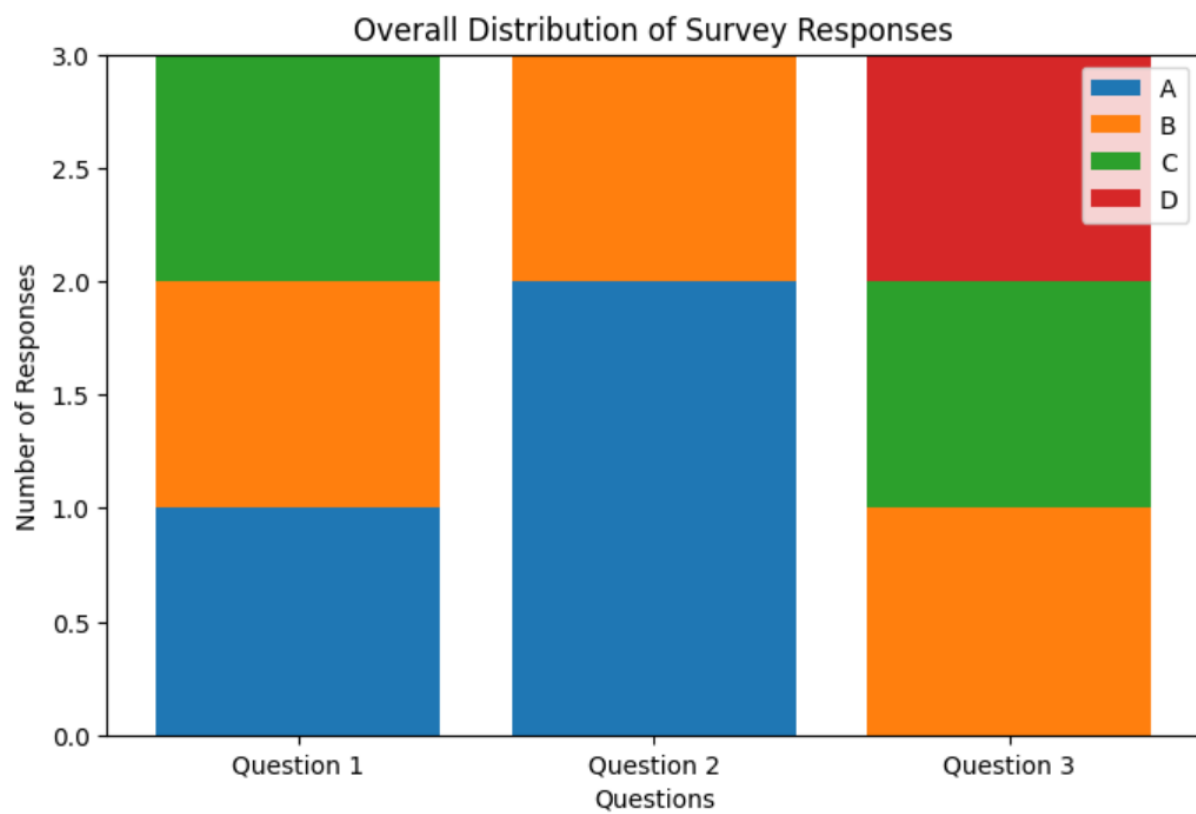
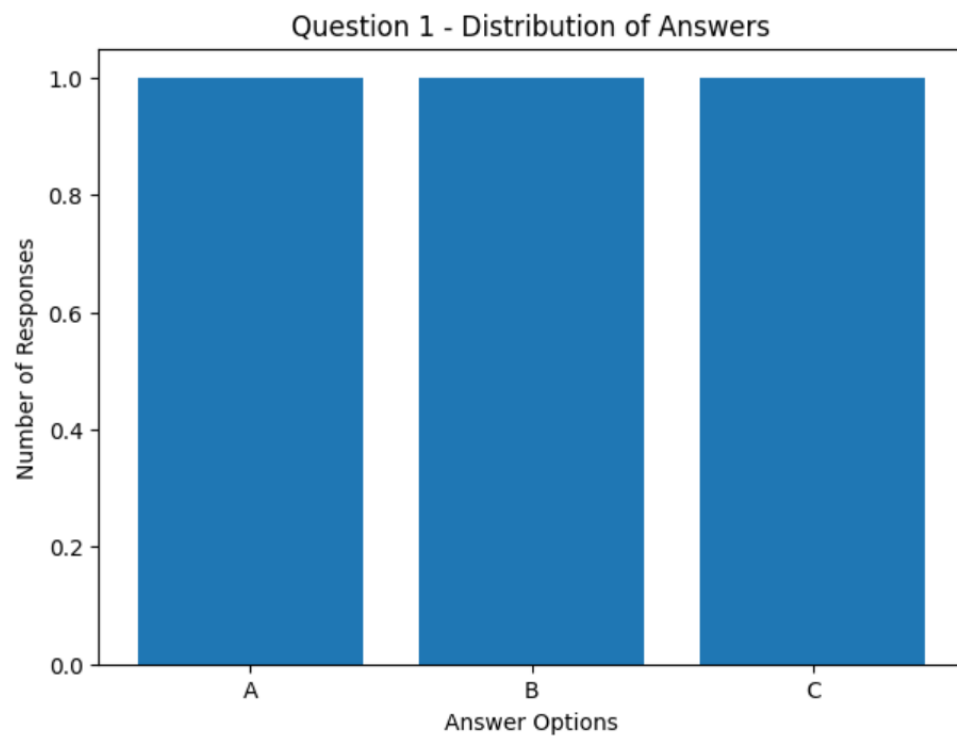
        "\t\t",

        q3[i]

    )
```

```
print("-" * 55)
```

OUTPUT:



SURVEY RESPONSE TABLE

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B